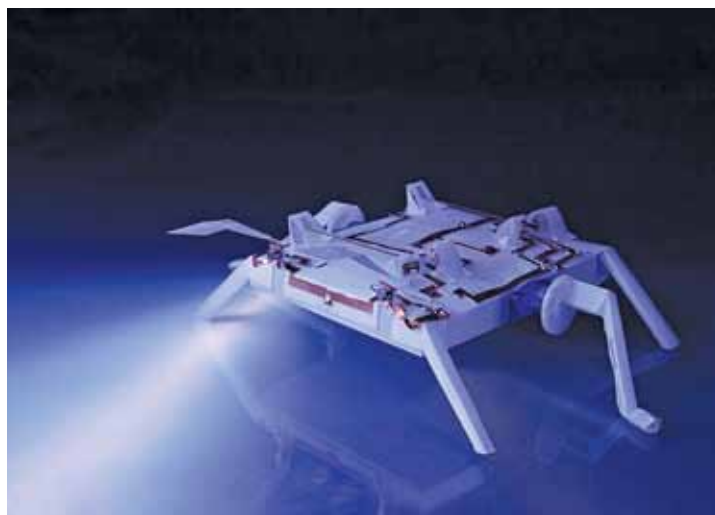


Flexible foldable robots that can perform variety of complex tasks

A multidisciplinary team led by researchers at the UCLA Samueli School of Engineering has created a new fabrication technique for fully foldable robots that can perform a variety of complex tasks without relying on semiconductors. Flexible robots, named Origamechs, created from pre-cut polyester sheets embedded with conductive materials, programmed with computer functions, and assembled into transistors with sensors and actuators, can sense, analyse, and act with precision. Origamechs derive computing power from origami switches made by folds and programmed Boolean logic commands, such as AND, OR, and NOT. These switches output electrical signals selectively based on variable pressure and heat input. The team built an insect-like walking robot, a Venus flytrap-like robot, and a reprogrammable two-wheeled robot to demonstrate the system's potential.



An origami-inspired robot designed by an UCLA-led team that can reverse direction when either of its antennae senses an obstacle (Credit: Wenzhong Yan/UCLA)

New system allows 'touching' and manipulation of holograms

Researchers at University of Calgary have introduced Holo-Touch, a system that can augment mixed reality graphics and charts using smartphones as physical proxies. The system combines HMDs and touchscreens on smartphones

or tablets to enhance MR visualisations. Essentially, Holo-Touch allows users to 'touch' and manipulate holograms that they are seeing on HMDs through their smartphone devices. Using a prototype, researchers showcased six ways

of interacting with visualised data. Preliminary evaluations revealed that the proxy improved people's exploration experience, with precision and tangibility as its strong suits, indicating that physical interaction with holograms is advantageous for data analysis tasks.



HoloTouch is an interactive mixed reality data visualisation that is built in two parts: (1) The handheld smartphone proxy that handles user interaction, and (2) a holographic visualisation that is rendered on top of the proxy. The user can create a selection by tapping the visual marks at its base (orange), then explore that subset of data in the green area using various features (Credit: Chulpongsatorn, Willett and Suzuki)